

BeliefBench: A Probabilistic Measurement Framework for Evaluating Belief Elicitation in Large Language Models

Technical Report

Matthew Dixon

Quiota LLC — Technical Report, Version 1.0.0

July 10, 2026

Abstract

Large language models increasingly report probabilities that users may treat as beliefs, yet these reports are elicited outputs rather than direct observations of an internal state. Conventional accuracy and RAG evaluations cannot determine whether such probabilities are reliable, calibrated, stable, appropriately responsive to evidence, or coherent with subsequent actions. BeliefBench addresses this problem by generating evidence from a user-defined Bayesian network and scoring the LLM against the resulting exact posterior. In a controlled GPT-4.1-mini financial case study, elicited beliefs were comparatively accurate, calibrated, and repeatable, but only 57.5% of actions were optimal under the model’s own reported probabilities. The actionable conclusion is conditional: validated probability reports may support monitored downstream risk and decision systems, while direct model actions should be withheld when decision-coherence tests fail. BeliefBench gives practitioners the evidence needed to make that distinction rather than assigning a blanket trust label.

Contents

1	Introduction	2
1.1	Contributions	2
2	Benchmark Design Principles	3
2.1	Notation	3
3	Probabilistic Framework	3
3.1	Directed graphical model and exact inference	3
3.2	Information views	3
4	Benchmark Generator	4
5	Belief Elicitation as Statistical Measurement	4
6	Evaluation Framework	5
7	Why BeliefBench?	5
8	Benchmark Specification	6
8.1	World, prompts, and adapters	6
8.2	Reproducible execution and artifacts	6
8.3	Financial case study	7
9	Experiment 1: Reliability of Elicited and Inferred Beliefs	7
9.1	Questions and protocol	7
9.2	Results	7
9.3	Audit and replication instructions	8

10 Discussion	9
10.1 What the empirical experiment establishes	9
10.2 Elicited versus inferred belief	10
10.3 Information-set mismatch and RAG evaluation	10
10.4 Entropy, dynamics, and governance	10
11 Limitations	11
12 Future Research	11
13 Conclusion	12
14 Related Work	12
A Formal Identifiability Results	14
B Minimum Reporting Checklist	15
C Output-to-Figure Map	15

1 Introduction

LLMs increasingly support forecasting, diagnosis, retrieval, and policy decisions in settings where uncertainty matters as much as the model’s most-probable answer. Yet common evaluations reduce a response to correctness, exact match, preference, or aggregate task score. Such metrics cannot determine whether a model assigns sensible mass to alternatives, is calibrated over repeated trials, changes its belief proportionally to evidence, or chooses actions consistent with its own stated probabilities. They also confound retrieval failure with probabilistic reasoning and output-format effects.

An LLM’s stated probability vector is an *elicited belief*: a response produced under a particular prompt, interface, and decoding configuration. It is not direct access to a persistent internal probability distribution, and fluent expression does not establish reliability. The empirical questions are therefore whether elicited probabilities track the uncertainty warranted by evidence, remain calibrated and repeatable, agree with independent behavioral indicators, and support coherent actions. BeliefBench treats the report as a statistical measurement whose validity must be tested rather than assumed. Every scenario begins with a sampled latent state and observations from a fully specified directed graphical model. Exact variable elimination supplies the reference posterior for the precise information available in each view. The same evidence is rendered as controlled natural language and presented with a prior and task, allowing a model’s elicited distribution, inferred distribution, and action to be compared with the exact target. Random seeds, conditional probability tables (CPTs), prompts, generated data, and metrics are retained, so an experiment can be reproduced or audited without reconstructing hidden annotations.

The key design choice is epistemic control. Conventional retrieval-augmented generation (RAG) can improve factuality by retrieving relevant text [12, 15], but the relationship between retrieved documents and the true latent state is rarely known. BeliefBench instead creates evidence through a Bayesian network and uses retrieval only as an information-delivery operation. It therefore supports controlled omissions and contradictions while keeping the full-data reference fixed, enabling a practical decomposition of retrieval, reasoning, and elicitation effects.

1.1 Contributions

BeliefBench contributes: (i) a configurable seeded probabilistic world with user-defined latent states, observable nodes, CPTs, and evidence language; (ii) exact variable-elimination posteriors for full, partial, and noisy evidence views; (iii) paired elicited-belief, inferred-belief, and decision measurements; (iv) proper scoring, divergence, stability, entropy, information-gain, and utility metrics; (v) a similarity-retrieval diagnostic under matched information; (vi) balanced-entropy generation, scenario-clustered uncertainty intervals, resumable measurement checkpoints, and auditable artifacts; and (vii) an automated dashboard that communicates model strengths and governance gates without inventing an opaque trust score. The empirical study shows why the instrument matters: GPT-4.1-mini is relatively calibrated and repeatable in the declared world, yet its actions are frequently inconsistent with its own reported beliefs.

2 Benchmark Design Principles

The benchmark follows five principles. First, the data-generating process is explicit: no scenario, label, or posterior is manually invented after generation. Second, uncertainty is first-class: scenarios are stratified by posterior entropy and analyzed as distributions rather than only labels. Third, information sets are recorded: the full evidence, presented evidence, and corrupted evidence remain distinguishable. Fourth, measurement channels are separated: elicited JSON beliefs, beliefs inferred from outputs, and actions are not treated as interchangeable. Fifth, reproducibility includes more than a seed; it includes software version, configuration, model identifier, sampling parameters, prompts, raw responses, and exact reference quantities.

These principles prevent a common interpretive error. Agreement with a sampled hidden state measures realized predictive success, whereas agreement with $P(S \mid E)$ measures probabilistic reasoning conditional on available information. A well-calibrated model may rationally assign substantial mass to a state that was not sampled. BeliefBench preserves both quantities but makes the posterior the primary target for belief-quality analysis.

2.1 Notation

Table 1 collects the principal symbols. Superscript v denotes an information view and subscript t a sequential step.

Table 1: Notation used throughout the paper.

Symbol	Meaning
$G = (\mathcal{V}, \mathcal{E})$	User-specified directed acyclic Bayesian graph.
$S \in \mathcal{S}$	Latent state; $\mathcal{S} = \{s_1, \dots, s_K\}$ is user-defined.
$E = (E_1, \dots, E_p)$	Observable and intermediate evidence variables.
θ	Complete benchmark configuration: graph, CPTs, templates, utilities, and controls.
r	Master random seed; scenario-specific child seeds are deterministic functions of r .
E^{full}, E^v	Complete evidence and evidence presented under view v .
b^{full}, b^v	Exact posteriors $P_\theta(S \mid E^{\text{full}})$ and $P_\theta(S \mid E^v)$.
$q^v, \tilde{q}^v$	Elicited probability vector and behaviorally inferred belief.
$a, a^*(q)$	Observed action and expected-utility-optimal action under q .
$U(a, s)$	User-specified payoff for action a in state s .
$H(\cdot), D_{\text{JS}}(\cdot, \cdot)$	Shannon entropy and Jensen–Shannon divergence.
T_t	Exact Bayesian transition operator induced by evidence arriving at step t .

3 Probabilistic Framework

3.1 Directed graphical model and exact inference

Let $S \in \{s_1, \dots, s_K\}$ denote a user-defined latent state. Observable nodes, their outcome cardinalities, parents, CPTs, and natural-language templates are supplied in YAML. The graph may contain direct observations and dependent intermediate nodes; hence the engine is not restricted to a naive-Bayes factorization. For an illustrative graph containing primary nodes E_i and dependent nodes V, L, N , the joint distribution factorizes as

$$P(S, E) = P(S) \prod_{i \in \mathcal{P}} P(E_i \mid S) P(V \mid S, C) P(L \mid S, F) P(N \mid V, L), \quad (1)$$

where $\mathcal{P}$ indexes primary nodes and V, C, L, F, N illustrate dependent variables. Outcome cardinalities are arbitrary finite positive integers, and each CPT row is normalized in the user configuration. This factorization permits correlated downstream observations while retaining exact inference.

For observed information $E_{\text{obs}} = e$, BeliefBench computes

$$b(s) = P(S = s \mid e) = \frac{\sum_{E \setminus (\{S\} \cup E_{\text{obs}})} P(S = s, E)}{\sum_{s'} \sum_{E \setminus (\{S\} \cup E_{\text{obs}})} P(S = s', E)}. \quad (2)$$

The sums are evaluated by variable elimination rather than sampling, particle filtering, or MCMC. Hence numerical reference error is limited to floating-point arithmetic and can be tested independently of the LLM.

3.2 Information views

For each sampled world the generator constructs three views. The *reference view* contains all generated observations. The *partial view* omits one or more variables, mimicking imperfect retrieval. The *noisy view* replaces an observation with

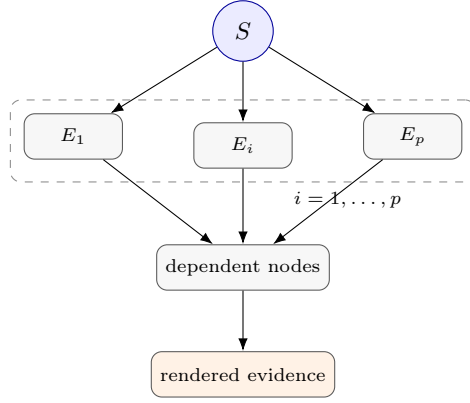


Figure 1: Bayesian graphical model in plate notation. The latent state and all node cardinalities, parents, and CPTs are user specified. The plate summarizes observable nodes; optional dependent nodes preserve correlations before categorical outcomes are rendered as evidence.

an irrelevant or contradictory value. Two posterior targets are retained: $b^{\text{full}} = P(S \mid E^{\text{full}})$ and the view-conditional posterior $b^v = P(S \mid E^v)$. The former anchors end-to-end loss from information mismatch; the latter isolates reasoning given exactly what the model saw.

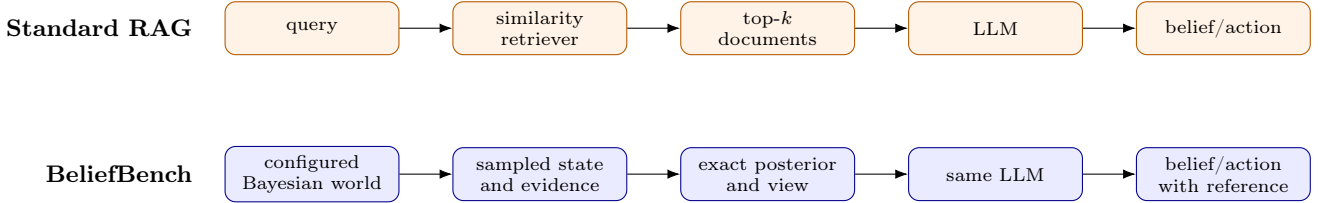


Figure 2: Standard RAG versus the BeliefBench workflow. Standard RAG retrieves text without a known probabilistic relationship to a latent state. BeliefBench generates evidence from a declared world and retains the exact posterior used for scoring.

4 Benchmark Generator

Given configuration θ and seed r , generation is deterministic: sample $S \sim P(S)$; sample nodes in topological order from their CPTs; calculate b^{full} and each b^v by exact elimination; convert categorical outcomes into templated natural-language evidence; create belief and decision prompts; and store the seed, latent state, evidence maps, posterior vectors, rendered text, and metadata. Natural-language templates vary surface expression without changing semantics. Because the categorical observation remains stored alongside its sentence, linguistic rendering can be audited separately from probabilistic generation.

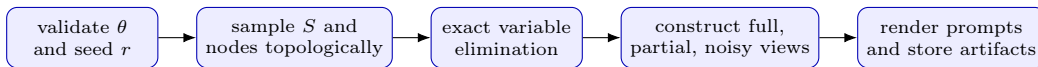


Figure 3: Benchmark generation pipeline. Each artifact is a deterministic function of the versioned configuration and seed except subsequent stochastic model calls.

The generator also creates sequential trajectories. At step t , the information set expands with a new observation and exact filtering gives $b_t = P(S \mid E_{1:t})$. This supports tests of update direction, information gain, and instability. The reference innovation can be measured as $D_{\text{JS}}(b_t, b_{t-1})$; model excess instability is then the divergence between a model update and the exact Bayesian transition, rather than merely movement between adjacent reports.

5 Belief Elicitation as Statistical Measurement

Let q_t be the normalized probability vector parsed from a model’s requested JSON response and $\tilde{q}_t$ a distribution inferred from an independent behavioral or scoring channel. BeliefBench deliberately avoids asserting $q_t = \tilde{q}_t$ or that either equals an inaccessible internal belief. Elicitation may be distorted by instruction following, quantization, verbal conventions, or decoding noise. Inference may be distorted by the chosen identification model. Their agreement is therefore an empirical validity test, not an assumption.

An independently elicited action $a_t \in \mathcal{A}$ provides a third channel. For payoff matrix $U(a, s)$, the action implied by

Configuration: DAG G , latent node S , CPTs θ , templates τ , seed r , number of scenarios n
Output: Scenarios $\mathcal{D} = \{(S_j, E_j, b_j^{\text{full}}, \{E_j^v, b_j^v\}_v)\}_{j=1}^n$
 Validate acyclicity, outcome cardinalities, CPT shapes and normalization, and template coverage;
 Compute a topological order π of G and spawn deterministic child seeds $r_1, \dots, r_n$;
for $j \leftarrow 1$ **to** n **do**
 Initialize assignment $x \leftarrow \emptyset$ and RNG with r_j ;
 foreach $X \in \pi$ **do**
 | sample $x_X \sim P_\theta(X \mid x_{\text{pa}(X)})$;
 end
 Set $E_j \leftarrow x_{V \setminus \{S\}}$ and compute $b_j^{\text{full}} \leftarrow P_\theta(S \mid E_j)$ by variable elimination;
 foreach *view* $v \in \{\text{reference, partial, noisy}\}$ **do**
 | Construct E_j^v by the declared view intervention;
 | Compute $b_j^v \leftarrow P_\theta(S \mid E_j^v)$ exactly and render E_j^v using τ ;
 end
 Persist $r_j, x, E_j, b_j^{\text{full}}, \{E_j^v, b_j^v\}_v$ and rendered evidence;
end

Algorithm 1. Exact Bayesian benchmark generation

belief q_t is

$$a^*(q_t) = \arg \max_{a \in \mathcal{A}} \sum_s q_t(s) U(a, s). \quad (3)$$

Decision consistency records $\mathbb{1}\{a_t = a^*(q_t)\}$, and reference agreement compares a_t with $a^*(b_t)$. Regret is

$$\max_a \mathbb{E}_{b_t} U(a, S) - \mathbb{E}_{b_t} U(a_t, S). \quad (4)$$

These measures distinguish coherent use of a possibly incorrect belief from a good belief used inconsistently.

6 Evaluation Framework

Posterior agreement is measured primarily by the bounded, symmetric Jensen–Shannon divergence

$$D_{\text{JS}}(q, b) = \frac{1}{2} D_{\text{KL}}(q \| m) + \frac{1}{2} D_{\text{KL}}(b \| m), \quad m = \frac{1}{2}(q + b), \quad (5)$$

supplemented by the multiclass Brier score

$$\text{BS}(q, S) = \sum_{s \in \mathcal{S}} (q_s - \mathbb{1}\{S = s\})^2, \quad (6)$$

log loss $-\log q_S$, and expected calibration error (ECE). Here q_s is the reported probability of state s , while $\mathbb{1}\{S = s\}$ equals one for the realized state and zero otherwise. The Brier score is therefore the squared distance between the probability vector and the one-hot realized outcome. Proper scores are included because probability forecasts must be rewarded for honest uncertainty rather than only top-class accuracy [2, 7]. Repeatability uses component-wise variance and mean pairwise JS divergence across identical calls. Entropy is $H(b) = -\sum_s b_s \log b_s$, information gain is $H(P(S)) - H(b)$, and entropy sensitivity relates $H(b)$ to repeated-call or sequential instability.

The mismatch design reports three descriptive components. Retrieval error is $D_{\text{JS}}(b^v, b^{\text{full}})$; reasoning error is $D_{\text{JS}}(q^v, b^v)$; total end-to-end error is $D_{\text{JS}}(q^v, b^{\text{full}})$. Because JS divergence is not additive, these quantities should not be presented as a literal algebraic identity without a residual or a path-based alternative. BeliefBench consequently stores all three and interprets them as complementary diagnostics.

7 Why BeliefBench?

BeliefBench adds a reference posterior and a known evidence mechanism to RAG evaluation. In ordinary corpora, a retrieved passage can be relevant yet stale, duplicated, dependent on another passage, or inconsistent with an unknown world state. Consequently, a model’s probability cannot be scored against a uniquely defined posterior. Here, relevance, omission, contradiction, dependency, and uncertainty are controlled by construction. Researchers can ask whether a model performs Bayesian updating, not merely whether it finds a plausible answer.

The API study demonstrates why this control is operationally useful. GPT-4.1-mini’s reference-view posterior error is 0.0789, but a top-state score alone records only 67.3% agreement with the exact posterior’s modal state and cannot reveal whether the remaining probability mass is sensible. BeliefBench additionally shows 5.3% expected calibration

error, low repeated-call instability (0.00536 mean pairwise JS), moderate elicited–inferred disagreement (0.0563 JS), and only 57.5% action consistency under the model’s own belief. These quantities answer different governance questions. They reveal that probability reports may be usable as monitored inputs even when the model’s direct actions should be rejected or recomputed by deterministic downstream code.

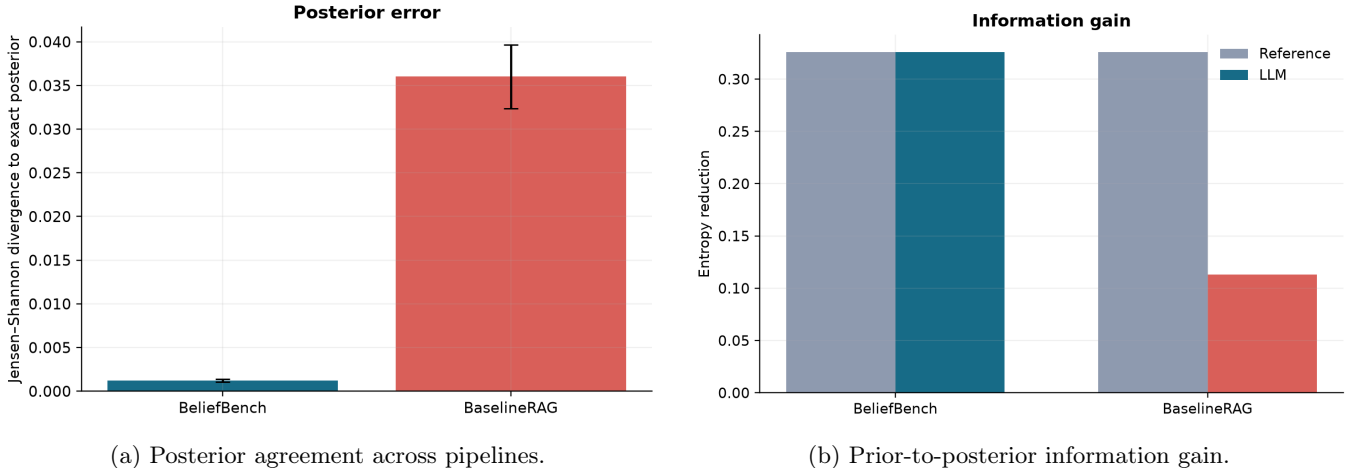


Figure 4: How to interpret the diagnostic comparison. The left panel reports divergence from the exact posterior, so shorter bars indicate more accurate elicited beliefs. The right panel reports the reduction from prior to posterior entropy, showing how much uncertainty each information pipeline removes. The panels measure complementary properties: posterior accuracy and information use.

The benchmark also yields matched evidence views and raw artifacts, which turns broad failure labels into testable hypotheses. For example, a model may be accurate under full information but brittle to one contradictory observation, stable yet systematically under-confident, or internally coherent while disagreeing with the exact posterior. Each pattern implies a different intervention and would be obscured by a single accuracy number.

8 Benchmark Specification

8.1 World, prompts, and adapters

The benchmark world formalizes the hypotheses that matter to the application and the observations that bear on them. The user chooses a mutually exclusive state space, a prior, and a directed graph whose conditional probability tables state how likely each observation is under each relevant parent configuration. This is the source of experimental ground truth: it determines both the evidence seen by the model and the exact posterior against which the elicited response is judged. State labels and cardinality are user-defined rather than financial or otherwise domain-specific.

Categorical observations are translated into short natural-language evidence statements. The prompt presents only the declared prior, the evidence available in that view, and the request for a probability distribution over the user-defined states. A separate decision prompt adds possible actions and their state-contingent utilities. This separation matters scientifically because it distinguishes three objects that might otherwise be conflated: the information available to the model, the probabilities the model reports, and the action it selects. Validation of the graph, probability tables, evidence coverage, and utility dimensions ensures that discrepancies can be attributed to model behavior rather than an incoherent experimental specification.

The BELIEFBENCH method receives evidence generated from the configured network with its probabilistic semantics preserved. The BASELINERAG comparison uses deterministic similarity retrieval over templates and domain-neutral distractors with no encoded link to the hidden state or posterior. Both feed the same downstream interface. The baseline is intentionally simple and should be supplemented by stronger sparse, dense, and reranked retrievers in substantive comparisons.

8.2 Reproducible execution and artifacts

Install the client-only wheel and run the hosted benchmark through the SDK:

```
python -m venv .venv
source .venv/bin/activate
pip install https://github.com/mfrdixon/BeliefBench-Python/releases/download/v0.6.0/beliefbench-0.6.0-py3-none-any.whl
```

```
export OPENAI_API_KEY='...'
export OPENAI_MODEL='gpt-4.1-mini'
```

The Python call is `BeliefBench().run("my_world.yaml")`. The configuration controls seeds, scenario count, channel-specific repeat counts, evidence corruption, model, checkpoint backend, and token ceilings. The hosted release currently supports OpenAI Responses API frontier models; the explicit SDK argument `openai_model`, the `OPENAI_MODEL` environment variable, and YAML permit model selection, while GPT-4.1-mini is the economical default. Complete installation, configuration, interpretation, and troubleshooting instructions are supplied in the versioned usage guide [3].

8.3 Financial case study

Finance is represented only as a separate example configuration. The file `examples/financial_full.yaml` defines the user-selected states Bull, Bear, and Crisis, prior (0.50, 0.30, 0.20), five financial observations and their CPTs, evidence templates, and utilities for Long, Neutral, and Cash. It is submitted with `BeliefBench().run()`; the same private engine runs arbitrary user-authored worlds. None of these financial labels occurs in prompt, inference, metric, or plotting logic.

Each run writes `summary.csv`, `belief_accuracy.csv`, `repeatability.csv`, `decision_consistency.csv`, `entropy_analysis.csv`, and trajectory records. It also writes posterior-error, repeatability, calibration, entropy-instability, decision, information-gain, trajectory, simplex, and executive-dashboard plots. The API study additionally writes per-seed ZIP archives, combined tables, entropy-balance audits, and scenario-clustered bootstrap intervals. Measurement-level checkpoints prevent successful paid calls from being discarded after transient failures.

9 Experiment 1: Reliability of Elicited and Inferred Beliefs

9.1 Questions and protocol

Experiment 1 asks whether GPT-4.1-mini produces accurate, calibrated, repeatable, behaviorally coherent, and decision-consistent beliefs; whether those properties degrade appropriately under information loss and corruption; and whether controlled evidence delivery improves on a diagnostic similarity retriever. Table 2 reports the complete design. Accept–reject sampling produced exactly 10 scenarios per entropy stratum per seed. Five elicitation repeats identify repeated-call instability, while inferred beliefs and decisions are measured once per condition to avoid paying repeatedly for channels whose repeatability is not the primary estimand. BaselineRAG is restricted to the reference view so the primary focus remains the absolute quality of the named model.

Table 2: Empirical API-study parameters.

Parameter	Value
Provider/model	OpenAI Responses API; gpt-4.1-mini
Latent states/prior	Bull, Bear, Crisis; (0.50, 0.30, 0.20)
Seeds	20260710, 20260711, 20260712, 20260713
Scenarios	30 per seed; 120 total
Entropy strata	10 low, 10 medium, 10 high per seed; cut points (0.45, 0.75)
Evidence views	Reference, partial (two omissions), noisy (one contradiction)
Methods	BeliefBench on all views; BaselineRAG on reference view
Elicitation repeats	5 per scenario–method–view condition
Inferred/action repeats	1 per condition; 12 binary payoff menus
Trajectories	10 steps per seed and method
Output ceiling	150 tokens per Responses API call
Concurrency/recovery	8 measurement workers per seed; 3 seed jobs; global 420 RPM pacing; durable measurement IDs; up to 8 seed attempts with 90-second cooldown
Inference	Exact variable elimination
Uncertainty intervals	2,000 bootstrap replications clustered by seed and scenario
Completed artifacts	2,400 elicitation rows; 480 decision rows; 120 unique scenarios

9.2 Results

Table 3 shows that controlled evidence delivery improves every matched reference-view error and decision metric. The result is not merely comparative: the absolute panel characterizes the model itself. ECE is 0.0533, mean pairwise repeated-call JS is 0.00536, and the model agrees with the exact posterior’s modal state in 67.3% of observations. Elicited and inferred beliefs differ by mean JS 0.0563, so behavioral evidence supports partial convergent validity but

Configuration: Generated scenarios $\mathcal{D}$, methods $\mathcal{M}$, views $\mathcal{V}$, repeat count R , LLM configuration $\mathcal{C}$, utilities U

Output: Belief, stability, calibration, mismatch, and decision records

```

foreach scenario  $j \in \mathcal{D}$  do
  foreach view  $v \in \mathcal{V}$  do
    Retrieve stored exact targets  $b_j^{\text{full}}$  and  $b_j^v$ ;
    foreach method  $m \in \mathcal{M}$  do
      Construct method-specific documents while holding the downstream LLM configuration fixed;
      for  $r \leftarrow 1$  to  $R$  do
        Elicit  $q_{jmv}$  and checkpoint the successful measurement under a deterministic ID;
        if  $r = 1$  then
          | independently estimate  $\tilde{q}_{jmv}$  and elicit action  $a_{jmv}$ ;
        end
        Record  $D_{\text{JS}}(q_{jmv}, b_j^{\text{full}})$ ,  $D_{\text{JS}}(q_{jmv}, b_j^v)$ , proper scores, and available action diagnostics;
      end
      Compute within-scenario component variance and mean pairwise JS across repeats;
    end
  end
end

```

Aggregate by prespecified method, view, and entropy stratum; retain scenario-level rows and raw responses;

Algorithm 2. Experimental protocol

Table 3: Matched reference-view GPT-4.1-mini results. Lower is better except agreement.

Method	Elicited JS	Inferred JS	Brier	Repeat. var.	Ref. action	Regret
BELIEFBENCH	0.0789	0.1370	0.1650	0.00207	0.700	0.0496
BASELINERAG	0.1238	0.1548	0.2136	0.00312	0.450	0.1812

not channel equivalence. The model’s action is optimal under its own elicited probability vector in only 57.5% of reference-view conditions, a failure invisible to ordinary answer-accuracy evaluation.

The information-set experiment behaves sensibly: mean total JS increases from 0.0789 on reference evidence to 0.0949 under omission and 0.1143 under contradiction. Mean retrieval mismatch is 0, 0.0653, and 0.0855, while view-conditional reasoning/elicitation error is 0.0789, 0.0594, and 0.0946. These quantities are descriptive rather than additive, but they show why a single end-to-end error cannot distinguish missing evidence from model processing.

Table 4: BeliefBench reference-view results by balanced posterior-entropy stratum (200 elicitation rows per stratum).

Entropy stratum	Mean JS	Brier	Log loss
Low	0.0943	0.0743	0.4419
Medium	0.0584	0.1976	0.9289
High	0.0840	0.2232	1.0976

The non-monotone JS pattern shows that posterior entropy is not identical to belief error. In particular, low-entropy scenarios can expose under-sharpening or misplaced residual mass even when the modal state is obvious. Brier score and log loss rise with entropy partly because the realized-state prediction problem is intrinsically harder. Mechanism claims about entropy and instability still require regressions that control for information gain, evidence conflict, view, model, time step, and scenario effects. Both $D_{\text{JS}}(q_t, q_{t-1})$ and transition-adjusted $D_{\text{JS}}(q_t, Tq_{t-1})$ should be reported in sequential studies.

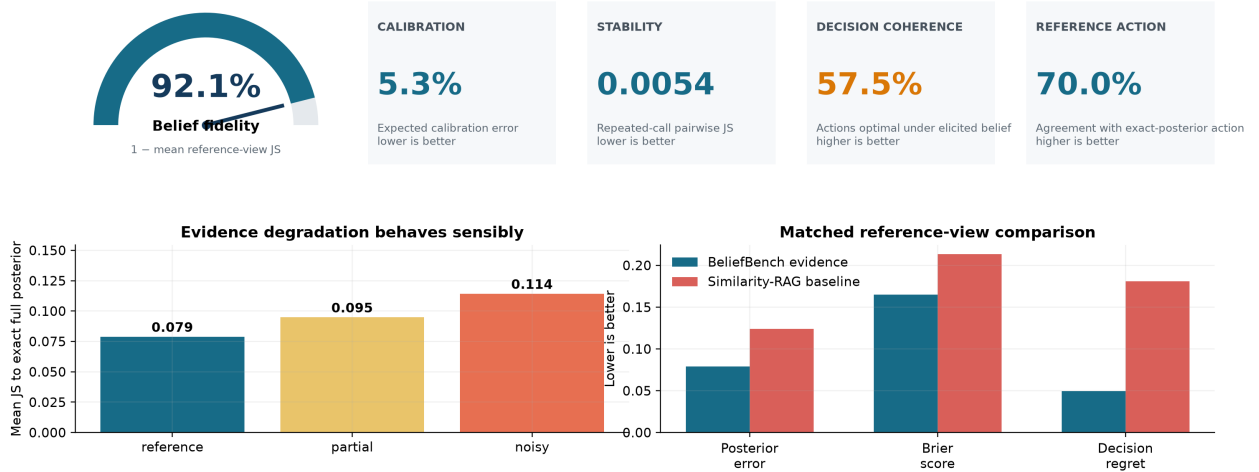
9.3 Audit and replication instructions

Replication requires preserving the probabilistic experiment and the measurement conditions, not merely rerunning a model name. The graph, CPTs, prompts, seed list, model identifier, execution date, and analysis choices jointly define the estimand. Provider refusals, parse failures, retries, missing observations, and interrupted calls must be reported because selective omission can make a model appear more reliable than it is. Likewise, uncertainty intervals should preserve the scenario as the clustering unit: repeated prompts for one scenario are not independent new worlds.

The completed study provides a concrete audit target. It contains 120 unique scenarios, exactly 40 in each posterior-entropy stratum, 2,400 elicitation measurements, and 480 decision measurements, with no duplicated measurement identifiers. A successful replication should recover these design counts before comparing substantive estimates. It should also record the API date and full model identifier, since behavior may change even when a provider retains the same public-facing model name. These requirements protect the scientific interpretation of the results; they are not incidental software bookkeeping.

BeliefBench · Frontier-model belief diagnostics

Exact-posterior evaluation of accuracy, calibration, stability, evidence robustness, and decisions



Interpretation: strong aggregate diagnostics support conditional use of elicited probabilities; decision consistency remains a governance gate, not a blanket trust score.

Figure 5: Empirical BeliefBench dashboard. The 92.1% dial is 1–mean reference-view JS and is not a universal trust score. The dashboard deliberately juxtaposes favorable accuracy, calibration, and stability with the weaker 57.5% decision-coherence result.

Table 5: Paired BeliefBench-minus-BaselineRAG effects on the matched reference view. Intervals are 95% scenario-clustered bootstrap intervals; negative is favorable for error and regret.

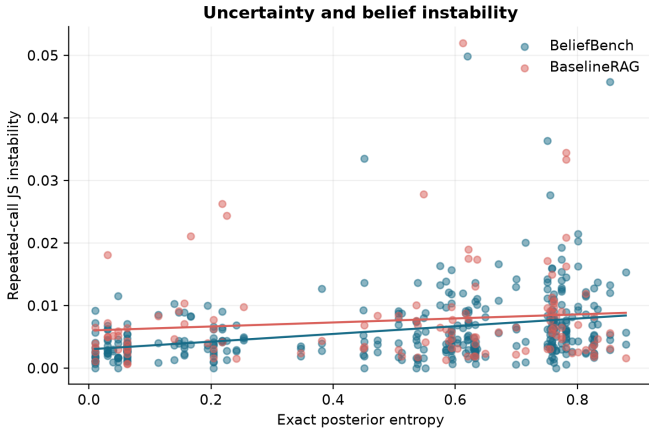
Estimand	Difference	95% interval
JS posterior error	−0.0449	[−0.0582, −0.0326]
Brier score	−0.0485	[−0.0638, −0.0327]
Log loss	−0.2415	[−0.3271, −0.1621]
Decision consistency	+0.1417	[+0.0415, +0.2500]
Reference-action agreement	+0.2500	[+0.1500, +0.3500]
Decision regret	−0.1316	[−0.1803, −0.0854]

10 Discussion

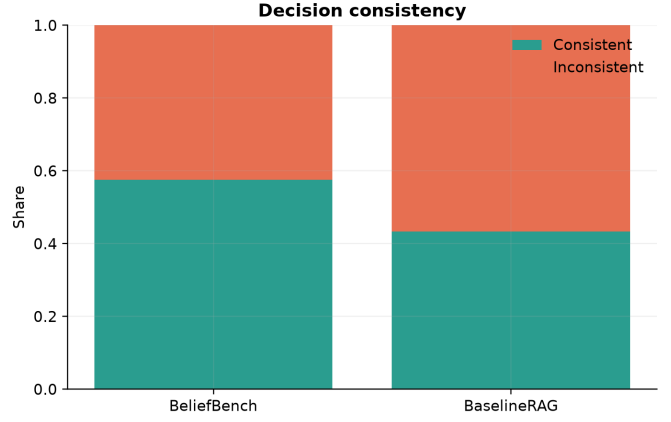
10.1 What the empirical experiment establishes

The experiment establishes a bounded empirical profile of GPT-4.1-mini’s observable belief interface in the declared financial world. Its elicited probabilities are reasonably close to the exact posterior, aggregate calibration error is modest, repeated calls are stable, and corruption degrades performance in the expected order. At the same time, modal-posterior agreement is only 67.3%, inferred beliefs are not identical to elicited beliefs, and 42.5% of reference-view actions are inconsistent with the model’s own probability report. BeliefBench therefore changes the governance decision from an unsupported binary claim that the model is or is not “trustworthy” to a conditional claim about which interface components meet a declared threshold.

The practical value is architectural. If belief accuracy, calibration, and stability pass application-specific gates while decision consistency fails, a system can retain the probability report but reject direct model actions. Deterministic downstream software can apply a validated utility matrix, trigger human review near decision boundaries, abstain under high entropy, or alert when elicited and inferred channels diverge. Conventional correctness benchmarks cannot support this selective deployment decision because they do not expose the full distribution or its relationship to action. The results also illustrate why posterior agreement is more informative than classification accuracy. A top-label score cannot distinguish (0.51, 0.48, 0.01) from (0.98, 0.01, 0.01) when both select the first configured state, despite radically different uncertainty and behavior near payoff boundaries. JS divergence, proper scores, calibration, and regret expose different aspects of that difference. No one statistic is sufficient; their joint pattern is the object of analysis.



(a) Entropy and repeated-call instability.



(b) Action consistency under elicited beliefs.

Figure 6: Uncertainty and decisions. The empirical design contains exactly 40 scenarios per entropy stratum. Entropy-stratified error is descriptive because the strata also differ in intrinsic prediction difficulty; decision consistency is a direct internal-coherence diagnostic conditional on the declared utilities.

10.2 Elicited versus inferred belief

BeliefBench’s distinction between q and $\tilde{q}$ is central. Explicit probabilities are influenced by response conventions, rounding, and prompts, while inferred beliefs depend on an observation model connecting language or choices to latent probabilities. Agreement supports convergent validity, but disagreement is not automatically evidence that one channel is “true.” It indicates that the measurement model requires investigation. Useful follow-ups include prompt paraphrases, proper-scoring incentives, token-level likelihood extraction where available, repeated binary bets, and held-out action matrices. A latent-variable model could combine these channels while estimating channel-specific bias and noise.

10.3 Information-set mismatch and RAG evaluation

The full/partial/noisy design creates a capability unavailable in most open-domain RAG benchmarks: the analyst knows both the complete probabilistic world and the information actually shown. Retrieval error can therefore be measured before reasoning begins. This matters because a seemingly irrational posterior may be appropriate for an incomplete view, while an accurate answer can arise from prior knowledge not contained in retrieved evidence. BeliefBench makes such leakage testable by using synthetic semantics and by comparing against both full- and view-conditional posteriors. The framework should not be interpreted as claiming that Bayesian RAG is universally superior to semantic retrieval. Real systems need both: semantic retrieval locates potentially useful observations, while a probabilistic layer represents reliability, dependence, and update structure. BeliefBench offers a controlled laboratory for testing that composition. Stronger baselines should attach calibrated likelihood models to dense retrieval scores, compare rerankers, and test whether probabilistic metadata improves decisions under identical context budgets.

10.4 Entropy, dynamics, and governance

Entropy is a property of the reference information state, not simply a prompt adjective. Balanced generation allows uncertainty to be manipulated while evidence length and semantic form remain controlled. Sequential observations further separate raw movement $D_{JS}(q_t, q_{t-1})$ from excess movement relative to the exact update $D_{JS}(q_t, Tq_{t-1})$. The first captures total instability; the second asks whether the model moves beyond what new evidence warrants. Their distinction is essential whenever evidence itself changes.

The early case studies already provide two substantive lessons about information and posterior dynamics. First, removing or corrupting evidence increases posterior error in the expected order, demonstrating that information quality is measurable as a graded influence on belief quality rather than a binary retrieval success. Second, error is not monotone in posterior entropy: low-entropy cases can still reveal under-sharpening or misplaced residual probability, while higher-entropy cases make realized-state prediction intrinsically harder. Entropy therefore measures ambiguity in the reference information state, not model error itself. In sequential work, the relevant scientific question is whether ambiguity changes the propagation of deviations after accounting for the Bayesian update warranted by new evidence. These distinctions matter directly in trading and risk management. Portfolio exposure depends on the entire regime distribution, especially residual crisis probability, rather than only the most likely Bull/Bear/Crisis label. A model that is stable but systematically underweights tails may support the wrong position size; a model that reacts excessively

to noisy news may generate turnover and transaction costs; and a model whose uncertainty rises appropriately may instead justify abstention, smaller risk limits, or human review. Belief-state filtering, model validation, and dynamic delegation provide complementary ways to translate these measurements into risk controls [4–6]. BeliefBench supplies the controlled evidence needed to test those mechanisms before they are embedded in a live trading policy.

For governance, a runtime statistic should be justified by an empirically validated mechanism. Belief instability or a belief-at-risk measure can be operationally useful, but it should not be treated as causal evidence by itself. A confirmatory design can test whether ambiguity amplifies endogenous propagation by estimating a lagged entropy-by-deviation interaction while controlling for current evidence, evidence shocks, chain effects, and time-step effects. BeliefBench supplies the controlled posterior and evidence innovations needed to identify that interaction more cleanly than free-form scenarios.

11 Limitations

Single model and provider. The empirical evidence concerns GPT-4.1-mini through the OpenAI Responses API at the recorded execution date. It does not establish the behavior of other OpenAI snapshots, providers, local models, or future deployments. The earlier simulated-adaptor results remain software tests and are not mixed with the empirical estimates.

External validity. The shipped domain-neutral world and the separate financial case study are intentionally small. Exact inference and interpretability are benefits, but natural domains contain continuous states, nonstationarity, feedback, strategic communication, and ambiguous ontology. Results do not transfer across domains without an appropriate user-specified network and validation.

CPT dependence. The reference posterior is exact conditional on the specified graph and CPTs, not metaphysically true. Poorly chosen probabilities can make a perfectly implemented benchmark unrepresentative. Sensitivity analysis should vary priors, likelihood ratios, graph density, and CPT concentration; domain experts should review semantics; and benchmark releases should version every parameter.

Retrieval baseline. The standard-RAG adaptor is a deliberately simplified comparator rather than a state-of-the-art dense retrieval system. Its documents lack probabilistic semantics by design, but this also makes the comparison asymmetric. Future studies should include BM25, dense passage retrieval, hybrid search, reranking, context compression, and retrieval-confidence calibration under matched token budgets.

Measurement identification. Neither elicited nor inferred probabilities provide direct access to an LLM’s internal state. Token probabilities may reflect surface realization; choices depend on payoffs and framing; repeated prompts may be correlated; and normalization can conceal malformed outputs. Multiple measurement channels, randomized prompt forms, and explicit missing-data rules are required.

Entropy and causal claims. The empirical run is balanced at 40 scenarios per entropy stratum, removing the earlier support imbalance, but balance alone does not identify a causal entropy effect. Entropy may correlate with evidence conflict, posterior margin, likelihood concentration, or information gain. The non-monotone JS pattern is descriptive; causal language requires randomized interventions or strong design assumptions and controls.

Metric limitations. JS divergence is bounded and symmetric but not additive, so retrieval, reasoning, and elicitation errors do not form a literal decomposition without a residual. ECE is bin-dependent and can look favorable in small samples [8, 20]. Log loss is sensitive to near-zero mass, while Brier score weights classes differently from decision regret. The benchmark therefore reports a metric panel and should include uncertainty intervals rather than ranking systems by a single decimal.

Sequential dependence and API drift. Trajectory observations share a latent state and repeated calls share prompts, invalidating iid standard errors. API models, system prompts, safety layers, and sampling implementations can change without a new public name. Empirical reports must use hierarchical or cluster-robust inference, timestamp calls, record deployment identifiers where possible, and preserve responses for audit.

Decision specification. Decision consistency is conditional on the supplied payoff matrix. Different utilities can reverse an optimal action without changing beliefs, and real users have heterogeneous risk preferences and constraints. Payoff sensitivity and multiple held-out matrices are needed before interpreting action agreement as broad rationality.

12 Future Research

Immediate work should replicate the balanced design across multiple model families, model sizes, prompt forms, and independently authored Bayesian worlds. A hierarchical measurement model could estimate a latent belief distribution from elicitation, token scores, bets, and actions while allowing channel-specific bias. Sequential extensions should use hidden Markov models or dynamic Bayesian networks with exact filtering where tractable and controlled approximations

otherwise.

The information-set experiment can be expanded from random omission to retrieval mechanisms with known recall, likelihood-ratio corruption, correlated duplicates, and adversarial contradictions. Causal designs can randomize evidence reliability and ambiguity while holding reference information gain fixed. Governance research can then derive thresholds from decision regret or transition-adjusted instability rather than selecting them heuristically. Finally, a community benchmark should add independently authored networks, signed dataset manifests, standardized API provenance, and a held-out evaluation server to reduce overfitting to the default graph.

13 Conclusion

BeliefBench addresses a practical gap between model capability and responsible use. An LLM can produce a plausible answer or an articulate probability vector without demonstrating that the probability is warranted by the available evidence. Because elicited beliefs are measurements produced through an interface, their accuracy, calibration, stability, evidence sensitivity, behavioral agreement, and decision coherence must be validated separately. The exact posterior supplied by a controlled Bayesian world makes those questions empirically answerable.

The GPT-4.1-mini case study shows why the distinction changes deployment decisions. The model’s probability reports were comparatively accurate, calibrated, and repeatable, and performance degraded sensibly as evidence was removed or corrupted. Nevertheless, only 57.5% of actions were optimal under its own elicited beliefs. A conventional answer score could not reveal this separation. For the studied configuration, the defensible action is to continue validating the probability interface while computing portfolio actions in deterministic, independently tested code and escalating uncertain or unstable cases for review.

Users should employ BeliefBench to make bounded decisions rather than to seek a universal trust score. Before deployment, they should define the relevant latent states and consequences, prespecify acceptance thresholds, test held-out worlds and evidence perturbations, compare elicited probabilities with behavioral and action channels, and retain uncertainty intervals across scenarios and seeds. A model that passes these gates may provide monitored probabilities for risk limits, scenario weighting, abstention, human-review triggers, or expected-utility calculations. A model that fails a gate points to a specific response: improve evidence delivery, recalibrate probabilities, constrain action authority, or reject the proposed use.

The current results are an initial characterization, not a final certification. Replication across models, prompts, retrievers, and independently authored worlds is required, while sequential experiments should test how entropy and information shocks affect raw and transition-adjusted belief dynamics. By retaining the hidden state, evidence, exact posterior, elicited response, action, and seed in one auditable chain, BeliefBench turns vague claims about model confidence into falsifiable evidence. That is the reason to use the benchmark: it supports a reasoned decision about what part of an LLM system, under what conditions, can be trusted with what level of authority.

14 Related Work

BeliefBench connects probabilistic graphical models, forecast evaluation, calibration, LLM benchmarking, uncertainty estimation, and RAG. Bayesian networks provide a compact factorization of joint distributions and support exact variable elimination under tractable graph structure [13, 18]. Proper scoring rules supply the normative basis for evaluating probability forecasts [2, 7], while modern calibration research shows that predictive accuracy does not guarantee reliable probabilities [8, 20].

LLM benchmarks such as MMLU, BIG-bench, HELM, and TruthfulQA broaden task and robustness coverage [1, 9, 16, 17], but primarily organize performance around task outcomes. Work on language-model calibration directly studies confidence quality and shows that generative probabilities can be miscalibrated in question answering [10, 21]. Other studies elicit verbal or numerical confidence, find that models can often predict whether their answers are correct, and emphasize prompt dependence [11, 19]. Semantic-entropy methods aggregate meaning-equivalent generations to detect uncertainty without relying on surface-form entropy [14]; BeliefBench is complementary because it supplies a known reference posterior and controlled evidence rather than estimating uncertainty solely from generations.

RAG combines parametric generation with retrieved nonparametric evidence [15]; dense passage retrieval and related retrievers made open-domain evidence acquisition effective at scale [12]. Subsequent evaluation frequently emphasizes answer quality, attribution, or retrieval relevance, whereas probabilistic semantics linking documents to a latent state are usually unavailable. BeliefBench changes the evaluation object: the retrieved text is generated from a known graphical model, and omissions or contradictions are interventions on a recorded information set. This permits an exact view-conditional posterior and a controlled comparison between retrieval and reasoning effects.

Finally, the benchmark draws on measurement theory’s distinction between a latent construct and an observed indicator.

Explicit probabilities, token likelihoods, repeated samples, and decisions are different indicators with different error mechanisms. BeliefBench operationalizes convergent validity by recording them separately and comparing their implications under known utilities. This design complements capability and factuality benchmarks rather than replacing them: it provides a probabilistic microscope for a narrower but consequential question—what can observable model behavior justify us in saying about belief under uncertainty?

The framework also complements recent work on agentic AI risk and governance. Belief-at-risk formalizes how inferred regime distributions and Bayesian filtering can support uncertainty and tail-risk measurement [4]; POMDP-based validation separates belief, forecast, policy, and utility risks [6]; and adaptive delegation treats decision authority as a sequential choice that should contract or expand with evidence and uncertainty [5]. BeliefBench operates upstream of these mechanisms by testing whether the observable probability interface is sufficiently well characterized to serve as an input to filtering, validation, or delegation.

References

- [1] BIG-bench collaboration. Beyond the imitation game: Quantifying and extrapolating the capabilities of language models. *Transactions on Machine Learning Research*, 2023.
- [2] Glenn W. Brier. Verification of forecasts expressed in terms of probability. *Monthly Weather Review*, 78(1):1–3, 1950. doi: 10.1175/1520-0493(1950)078<0001:VOFEIT>2.0.CO;2.
- [3] Matthew Dixon. *BeliefBench Usage Guide: Installation, Configuration, Experiments, and Interpretation*. Quiota LLC, 2026. URL <https://github.com/mfrdixon/BeliefBench/blob/main/docs/BeliefBench-Usage-Guide.pdf>. Version 0.1.0; software documentation.
- [4] Matthew Francis Dixon. Belief at risk: Quantifying agentic ai model risk with llm-inferred bayesian state filters. *arXiv preprint arXiv:2606.15473*, 2026. URL <https://arxiv.org/abs/2606.15473>.
- [5] Matthew Francis Dixon. Adaptive ai delegation under uncertainty: A bayesian governance policy for sequential decision authority. *arXiv preprint arXiv:2606.29406*, 2026. URL <https://arxiv.org/abs/2606.29406>.
- [6] Matthew Francis Dixon. Model validation of agentic ai systems: A pomdp-based framework for belief-state, forecast, and policy validation. *arXiv preprint arXiv:2606.17383*, 2026. URL <https://arxiv.org/abs/2606.17383>.
- [7] Tilmann Gneiting and Adrian E. Raftery. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359–378, 2007. doi: 10.1198/016214506000001437.
- [8] Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. In *Proceedings of the 34th International Conference on Machine Learning*, volume 70 of *PMLR*, pages 1321–1330, 2017. URL <https://proceedings.mlr.press/v70/guo17a.html>.
- [9] Dan Hendrycks, Collin Burns, Steven Basart, Andy Zou, Mantas Mazeika, Dawn Song, and Jacob Steinhardt. Measuring massive multitask language understanding. *Proceedings of ICLR*, 2021.
- [10] Zhengbao Jiang, Jun Araki, Haibo Ding, and Graham Neubig. How can we know when language models know? on the calibration of language models for question answering. *Transactions of the Association for Computational Linguistics*, 9:962–977, 2021. doi: 10.1162/tacl_a_00407.
- [11] Saurav Kadavath, Tom Conerly, Amanda Askell, et al. Language models (mostly) know what they know. *arXiv preprint arXiv:2207.05221*, 2022.
- [12] Vladimir Karpukhin, Barlas Oguz, Sewon Min, Patrick Lewis, Ledell Wu, Sergey Edunov, Danqi Chen, and Wen-tau Yih. Dense passage retrieval for open-domain question answering. In *Proceedings of EMNLP*, pages 6769–6781, 2020.
- [13] Daphne Koller and Nir Friedman. *Probabilistic Graphical Models: Principles and Techniques*. MIT Press, 2009.
- [14] Lorenz Kuhn, Yarin Gal, and Sebastian Farquhar. Semantic uncertainty: Linguistic invariances for uncertainty estimation in natural language generation. In *Proceedings of ICLR*, 2023.
- [15] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. Retrieval-augmented generation for knowledge-intensive nlp tasks. In *Advances in Neural Information Processing Systems*, volume 33, 2020.

- [16] Percy Liang, Rishi Bommasani, Tony Lee, et al. Holistic evaluation of language models. *Transactions on Machine Learning Research*, 2023.
- [17] Stephanie Lin, Jacob Hilton, and Owain Evans. Truthfulqa: Measuring how models mimic human falsehoods. In *Proceedings of ACL*, pages 3214–3252, 2022.
- [18] Judea Pearl. *Probabilistic Reasoning in Intelligent Systems*. Morgan Kaufmann, 1988.
- [19] Katherine Tian, Eric Mitchell, Allan Zhou, Archit Sharma, Rafael Rafailov, Huaxiu Yao, Chelsea Finn, and Christopher D. Manning. Just ask for calibration: Strategies for eliciting calibrated confidence scores from language models fine-tuned with human feedback. In *Proceedings of EMNLP*, pages 5433–5442, 2023.
- [20] Juozas Vaicenavicius, David Widmann, Carl Andersson, Fredrik Lindsten, Jacob Roll, and Thomas B. Schön. Evaluating model calibration in classification. In *Proceedings of the 22nd International Conference on Artificial Intelligence and Statistics*, volume 89 of *PMLR*, pages 3459–3467, 2019.
- [21] Shujian Zhang, Chengyue Gong, and Eunsol Choi. Knowing more about questions can help: Improving calibration in question answering. In *Findings of ACL-IJCNLP*, pages 1958–1970, 2021.

A Formal Identifiability Results

This appendix distinguishes identification of the benchmark reference posterior from identification of an LLM’s internal belief. The former follows from the declared probabilistic world; the latter requires additional measurement assumptions and is not claimed by construction.

Assumption 1 (Finite configured world). *$G = (\mathcal{V}, \mathcal{E})$ is a finite directed acyclic graph containing a designated latent node S with finite state space $\mathcal{S}$. Each node $X \in \mathcal{V}$ has a finite nonempty outcome space $\mathcal{X}_X$ and a complete CPT $\theta_X(x \mid x_{\text{pa}(X)})$ that is nonnegative and sums to one for every parent assignment.*

Assumption 2 (Positive observed event). *For the presented evidence event $E_O = e_O$, the configured model assigns $P_\theta(E_O = e_O) > 0$.*

Lemma 1 (Unique joint distribution). *Under Assumption 1, the graph and CPT collection θ induce a unique joint probability mass function*

$$p_\theta(x) = \prod_{X \in \mathcal{V}} \theta_X(x_X \mid x_{\text{pa}(X)}).$$

Proof. Because G is acyclic, it admits a topological order $X_1, \dots, X_d$. The chain rule gives $p(x) = \prod_{i=1}^d p(x_i \mid x_{1:i-1})$. The local Markov property encoded by G replaces each conditional by the configured $\theta_{X_i}(x_i \mid x_{\text{pa}(X_i)})$. Completeness fixes every factor for every assignment, so the product is unique and nonnegative. Summing successively in reverse topological order eliminates one normalized CPT at a time and yields $\sum_x p_\theta(x) = 1$. Thus the product is a unique probability mass function. $\square$

Theorem 1 (Reference-posterior identifiability). *Under Assumptions 1–2, the benchmark reference posterior $b^v(s) = P_\theta(S = s \mid E_O = e_O)$ is point identified for every $s \in \mathcal{S}$ and equals*

$$b^v(s) = \frac{\sum_{x_{\mathcal{V} \setminus (\{S\} \cup O)}} p_\theta(S = s, E_O = e_O, x)}{\sum_{s' \in \mathcal{S}} \sum_{x_{\mathcal{V} \setminus (\{S\} \cup O)}} p_\theta(S = s', E_O = e_O, x)}.$$

Proof. Lemma 1 uniquely fixes every joint atom. Finite marginalization therefore uniquely fixes the numerator $P_\theta(S = s, E_O = e_O)$ and denominator $P_\theta(E_O = e_O)$. Assumption 2 makes the denominator strictly positive, so division produces exactly one normalized posterior vector. Any alternative posterior compatible with the same G , θ , and observed event would have the same numerator and denominator for every state and must coincide componentwise with b^v . Hence the posterior is point identified. $\square$

Proposition 1 (View-specific identification). *If a partial or noisy view is a recorded deterministic or seeded stochastic transformation g_v of full evidence, and the realized presented assignment $e^v = g_v(e^{\text{full}})$ is stored, then both b^{full} and b^v are separately point identified whenever their evidence events have positive probability.*

Proof. Apply Theorem 1 once with the full observation index set and once with the index set and values actually presented. Recording the realized transformation removes ambiguity about the LLM’s information set. The two posteriors need not be equal, but each is unique under the same configured joint distribution. $\square$

Proposition 2 (Reproducibility of generated references). *Suppose the pseudorandom-number generator, seed-spawning rule, graph traversal order, and floating-point inference implementation are fixed. Then Algorithm 1 maps (θ, r, n) to a unique ordered dataset and collection of reference posterior vectors.*

Proof. The master seed deterministically produces child seeds. For each child seed, inverse-transform categorical sampling with the fixed generator selects one outcome at every node in the fixed topological order. View transformations are likewise fixed by stored child randomness. Theorem 1 makes every posterior unique for the resulting evidence. Composition of these deterministic maps gives the result. $\square$

Proposition 3 (Observable measurement distribution). *For a fixed scenario, prompt, deployment, and decoding configuration $\mathcal{C}$, repeated calls identify features of the observable response distribution $P(q \mid E, \mathcal{C})$ under independent, stationary sampling; they do not by themselves identify a unique internal LLM belief.*

Proof. Under independent stationary sampling, empirical means, component variances, and bounded pairwise-divergence averages converge to their corresponding response-distribution functionals by the law of large numbers. However, multiple latent internal-state and reporting-channel models can induce the same observable distribution over q . Without an injective measurement model or additional exclusion restrictions, these observationally equivalent latent representations cannot be distinguished. Thus observable reliability is identified, whereas internal belief is not generally point identified. $\square$

The positive-evidence condition is essential. If $P_\theta(E_O = e_O) = 0$, regular conditional probabilities on that null event are not determined by the finite ratio above. BeliefBench prevents such cases during ordinary generation because sampled assignments lie in the support, but a manually forced noisy view should be checked explicitly.

B Minimum Reporting Checklist

Report the repository commit; configuration and seed; graph and CPT version; model and API date; decoding parameters; prompt templates; scenario, view, and repeat counts; entropy-stratum counts; parse and refusal rates; all posterior targets; elicited and inferred channel definitions; utility matrices; confidence intervals with clustering unit; multiplicity policy; and links or hashes for raw artifacts. Clearly label simulated-adaptor, local-model, and remote-API results.

C Output-to-Figure Map

Posterior and calibration figures derive from `belief_accuracy.csv`; repeatability uses `repeatability.csv`; action plots use `decision_consistency.csv`; entropy plots use `entropy_analysis.csv`; and aggregate claims should be recoverable from `summary.csv`. To update manuscript figures, rerun the benchmark, verify row counts and adaptor labels, then copy the regenerated PNGs from `benchmark_results/plots/` into the manuscript’s `figures/` directory before compiling.